layer. This electric field is strong enough to break or rupture the covalent bonds in the lattice and thereby generating electron-hole pairs. This effect is called **Zener effect**.

Even a small further increase in reverse voltage produces a large number of charge carriers. Hence the junction has very low resistance in the breakdown region. This process of emission of electrons due to the rupture of covalent bonds in the lattice due to strong electric field is known as **internal field emission or field ionization**. The electric field required for this is of the order of 10^6 V m^{-1}.

ii) Avalanche breakdown

Avalanche breakdown occurs in lightly doped junctions which have wide depletion layers. In this case, the electric field is not strong enough to produce breakdown. Alternatively, the thermally generated minority charge carriers accelerated by the electric field gain sufficient kinetic energy and they collide with the semiconductor atoms while passing through the depletion region. This leads to the breaking up of covalent bonds generating electron-hole pairs.

The newly generated charge carriers are also accelerated by the electric field resulting in more collisions and further production of charge carriers. This cumulative process leads to an avalanche of charge carriers across the junction and consequently reduces the reverse resistance. The diode current increases sharply.

Note

For a reverse voltage of,

(i) less than 4V → Zener effect predominates

(ii) greater than 6V → Avalanche effect predominates

(iii) between 4 and 6V → both effects are present.

10.3.6 Zener diode

Zener diode is a heavily doped silicon diode used in reverse biased condition and is named after its inventor Clarence Melvin Zener. It is specially designed to be operated in the breakdown region. The doping level of the silicon diode can be varied to have a wide range of breakdown voltages from 2 V to over 1000 V.

As explained in the previous section, Zener breakdown occurs due to the breaking up of covalent bonds by the strong electric field set up in the depletion region by the reverse voltage. It produces an extremely large number of electrons and holes which constitute the reverse saturation current. The current is limited by both external resistance and power dissipation of the diode. A Zener diode is shown in Figure 10.19(a) and its circuit symbol is given in Figure 10.19(b).

It looks like an ordinary *p-n* junction diode except that *n*-side lead resembles the shape of the letter 'z'. The arrow head points the direction of conventional current. In Figure 10.19(a), black ring indicates the *n*-side lead.



Figure 10.19 Zener diode
(a) Commercial picture (b) Circuit symbol

V-I Characteristics of Zener diode

The circuit to study the forward and reverse characteristics of a Zener diode is shown in Figure 10.20(a) and Figure 10.20 (b). The V-I characteristics of a Zener diode is shown in Figure 10.20(c). The forward characteristic of a Zener diode is similar to that of an ordinary *p-n* junction diode. It starts conducting approximately around 0.7 V. However, the reverse characteristics is highly significant in Zener diode. The increase in reverse voltage

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